

Imputation of PEG genotype data to HRC reference panel

Install the latest version of:

Software for handling VCF files and indexing: BCFtools

Software suite for GWAS: PLINK2

Compression for bzip2 files: 7z

Prepare libraries in R that were referenced in PLINK2 tutorial:

```
packages <- c(
  "HardyWeinberg",
  "glue",
  "ggplot2",
  "plotly",
  "dplyr",
  "patchwork",
  "pROC",
  "ggforce",
  "data.table",
  "ggpubr",
  "ggrepel"
)

install_if_missing <- function(package) {
  if (!require(package, character.only = TRUE)) {
    cat(sprintf("Installing package: %s\n", package)) # Message indicating installation
    install.packages(package, dependencies = TRUE)
  } else {
    cat(sprintf("Package %s is already installed.\n", package)) # Message indicating already installed
  }
}

# Install the required packages
invisible(sapply(packages, install_if_missing))

# Confirm that installation is complete
cat("All packages checked and installed if missing.\n")
```

```
#source("../rawdata/R/loadplink2HW.R") #source("../rawdata/R/popstrat.R") #source("../rawdata/R/gwas1.R")
```

Download from the Box folder and save into ../rawdata/Genetics:

- PEG_PD.phased.vcf.gz
- gwas_covariates.tar.gz

Convert the original VCF file into a PLINK2 formatted set of files:

```
plink2 --vcf PEG_PD.phased.vcf.gz --make-pgen -out results/qc/qc1/PEG_PD_RAW
```

Then run the R script to generate a PLINK2 friendly pheno.txt by extracting data from the raw covariate files:

```
df_peg1_cov<-read.csv('peg1_covariates.csv')
df_peg1_cov<-df_peg1_cov[,c('ExternalDNACode','Female','Age','Ethnicity','PDstudyDiseaseNumeric')]
colnames(df_peg1_cov)<-c('PEGID','FEMALE','AGE','ETHNICITY','DISEASE')
df_peg1_cov$COHORT<-'PEG1'

df_peg2_cov<-read.csv('peg2_covariates.csv')
df_peg2_cov<-df_peg2_cov[,c('PEGID','Female','Age','EthnicityOriginal')]
df_peg2_cov$DISEASE<-1
colnames(df_peg2_cov)<-c('PEGID','FEMALE','AGE','ETHNICITY','DISEASE')
df_peg2_cov$COHORT<-'PEG2'

df_peg_all_cov<-rbind(df_peg1_cov,df_peg2_cov)
colnames(df_peg_all_cov)[4]<- 'ETHNICITY'
df_peg_all_cov$DISEASE<-df_peg_all_cov$DISEASE+1

df_linkage<-read.csv('gwas_linkage.csv')
colnames(df_linkage)[1]<- 'PEGID'

df_plink2_samples<-read.table('results/qc/qc1/PEG_PD_RAW.psam',sep='\t')
colnames(df_plink2_samples)<-c('GWAS_ID','SEX')
df_plink2_samples<-merge(df_plink2_samples,df_linkage,by=c('GWAS_ID'),sort=FALSE)
df_plink2_samples3<-merge(df_plink2_samples,df_peg_all_cov,by=c('PEGID'),sort=FALSE,all.x=TRUE)
df_plink2_samples3$FEMALE<-df_plink2_samples3$FEMALE+1
df_plink2_samples3$FEMALE[is.na(df_plink2_samples3$FEMALE)]<--9
df_plink2_samples3$AGE[is.na(df_plink2_samples3$AGE)]<--9
df_plink2_samples3$DISEASE[is.na(df_plink2_samples3$DISEASE)]<--9
df_plink2_samples3$ETHNICITY[df_plink2_samples3$ETHNICITY=='.']<- 'Unknown'
df_plink2_samples3$ETHNICITY[is.na(df_plink2_samples3$ETHNICITY)]<- 'NONE'
names(df_plink2_samples3)[which(names(df_plink2_samples3)=='GWAS_ID')]<- 'IID'
write.table(df_plink2_samples3[,c('IID','ETHNICITY','DISEASE')],file='pheno.txt',quote=F,row.names=F)
write.table(df_plink2_samples3[,c('IID','FEMALE','AGE','COHORT')],file='covar.txt',quote=F,row.names=F)
```

Filter the dataset to include only subjects with disease status:

```
plink2 --pfile results/qc/qc1/PEG_PD_RAW --pheno pheno.txt --make-pgen --require-pheno DISEASE -out res
```

Compute Hardy-Weinberg stats:

```
plink2 --pfile results/qc/qc1/PEG_PD_COV_FILT --hardy --snps-only --autosome --keep-founders --thin-ind
```

To get the HW Ternary plot in R that shows deviations across the 3 genotype classes:

```
# Load necessary package
library("HardyWeinberg")

# Function to load PLINK2 data into R
loadplink2Rmat <- function(file, header = TRUE, header_char = "#", sep = "\t", nrows = Inf) {
  # Read the header if it exists
  if (header) {
    header_line <- gsub(header_char, "", readLines(file, n = 1)) # Read and clean the header
    col_names <- strsplit(header_line, sep)[[1]] # Split header into column names
    data <- read.table(file, sep = sep, skip = 1, header = FALSE, nrows = nrows) # Read data, skipping
    colnames(data) <- col_names # Assign column names to the data
  } else {
```

```

    data <- read.table(file, sep = sep, header = FALSE, nrows = nrows) # Read the entire file without
    colnames(data) <- paste0("V", 1:ncol(data)) # Generate default column names
  }

  return(data) # Return the loaded data
}

fname = "results/qc/qc1/PEG_PD_HW_ALL_POP.hardy"

data<-loadplink2Rmat(fname)

HWTernaryPlot(data[,5:7],region=0,markercol = rgb(0,0,1,0.03),vertexlab = c("AA","AB","BB"))

```

QQ plot of HWE p-values:

```

format_hwddata2hwpack <- function(data) {
  x <- data[, 5:7] # Select relevant columns
  colnames(x) <- c("AA", "AB", "BB") # Set column names

  # Convert each column to numeric
  numeric_matrix <- do.call(cbind, lapply(x, function(column) as.numeric(unlist(column))))

  return(numeric_matrix) # Return the numeric matrix
}

fname = "results/qc/qc1/PEG_PD_HW_ALL_POP.hardy"

data<-loadplink2Rmat(fname)

data2 <- format_hwddata2hwpack(data)
HWQqplot(data2,nsim=10,logplot=TRUE, main="Observed QQ HWE-Null")

```

See the allele frequency spectrum:

```

plink2 --vcf PEG_PD.phased.vcf.gz \
  --freq alt1bins=0.01,0.02,0.03,0.04,0.05,0.06,0.07,0.08,0.09,0.10,0.11,0.12,0.13,0.14,0.15,0.16,0.17 \
  --snps-only \
  --autosome \
  --keep-founders \
  --thin-count 100000 \
  --out results/qc/qc1/PEG_PD_freq_ALLPOP_keepall

fname='results/qc/qc1/PEG_PD_freq_ALLPOP_keepall.afreq.alt1.bins'
data = loadplink2Rmat(fname)
barplot(unlist(data['OBS_CT']),names.arg=unlist(data['BIN_START']),ylim=c(0,1000),xlab="ALT1 FREQ",ylab="COUNT")

```

Remove SNPs with MAF <.01 call rates <.1 and out of HWE at $p < 1e-7$

```

plink2 --pfile results/qc/qc1/PEG_PD_COV_FILT --geno 0.1 --maf 0.01 -hwe 0.0000001 --export vcf --out :

```

Generate Chr bgzip files:

```

for chr in {1..22};
do
  echo Chrom $chr
  bcftools filter PEG_PD.phased.filtered.vcf.gz -r $chr > chunks/chr_$chr.vcf
  bgzip chunks/chr_$chr.vcf
done

```

```
tabix chunks/chr_${chr}.vcf.gz
done
```

Impute on imputation server after uploading these files.

Save results in ../rawdata/Genetics/imputed_raw:

```
# replace password below with yours
pwd='v18+UuqCENf>6P'
cd imputed_raw
for i in {1..22}
do
  echo "Chromosome $i"
  7z x -p$pwd chr_${i}.zip
done

# Merge chr vcfs into a single one.
ls chr*.dose.vcf.gz >filelist.txt
bcftools concat -f filelist.txt -O z -o chr_all.dose.vcf.gz
```

Convert back to plink2:

```
plink2 --vcf imputed_raw/chr_all.dose.vcf.gz dosage=DS --extract-if-info "R2 >= 0.8" --exclude-if-info
```

LD pruning:

```
plink2 --pfile results/qc/qc2/PEG_PD_IMPUTED --rm-dup force-first --make-pgen --out results/qc/qc2/PEG_
plink2 --pfile results/qc/qc2/PEG_PD_dedup --indep-pairwise 100 0.8 --out results/qc/qc3/PEG_PD_prune
```

Generate new dataset of independent variants:

```
plink2 --pfile results/qc/qc2/PEG_PD_dedup --make-pgen --extract results/qc/qc3/PEG_PD_prune.prune.in --
```

Compute Principal Components:

```
plink2 --pfile results/qc/qc3/PEG_PD_LE --freq counts --pca biallelic-var-wns --out results/qc/qc3/PEG_
sed 's/^#//' results/qc/qc3/PEG_PD_LE.psam > psam.tmp
sed 's/^#//' results/qc/qc3/PEG_PD_pca.eigenvec > evec.tmp
```

Generate PCA plots in R:

```
#source('../R/popstrat.R')
popstrat_plot2d <- function(A, PCID_, Aprefix = NULL, A2 = NULL, PCID_A2 = NULL, A2prefix = NULL, column

  A$selected<-A[,which(colnames(A)==column)]
  A2$selected<-A2[,which(colnames(A2)==column)]

  # Negate values based on PCID_ for A
  for (i in 1:2) {
    if (grepl("^-", PCID_[i])) {
      negate_col <- substring(PCID_[i], 2)
      A[[PCID_[i]]] <- -A[[negate_col]]
    }
  }

  A_names <- paste(Aprefix, A$selected)

  # Negate values for A2 if provided
```

```

if (!is.null(A2) && !is.null(PCID_A2)) {
  for (i in 1:2) {
    if (grepl("^-", PCID_A2[i])) {
      negate_col <- substring(PCID_A2[i], 2)
      A2[[PCID_A2[i]]] <- -A2[[negate_col]]
    }
  }
}

A2_names <- paste(A2prefix, A$selected)

# Create the 2D scatter plot using plotly
fig <- plot_ly() %>%
  add_trace(data = A, x = ~get(PCID_[1]), y = ~get(PCID_[2]), color = ~selected,
            colors = c('gold', 'red', 'green', 'blue', 'magenta'),
            type = "scatter", mode = "markers",
            marker = list(size = 8, opacity = 1),
            legendgroup = "A", name = A_names)

# Add A2 data to the plot if provided
if (!is.null(A2) && !is.null(PCID_A2)) {
  fig <- fig %>%
    add_trace(data = A2, x = ~get(PCID_A2[1]), y = ~get(PCID_A2[2]),
              type = "scatter", mode = "markers", color = ~selected,
              colors = c('gold', 'red', 'green', 'blue', 'magenta'),
              marker = list(symbol = "square", size = 8, opacity = 1,
                            line = list(color = "black", width = 1)),
              legendgroup = "A2", name = A2_names)
}

fig <- fig %>%
  layout(
    xaxis = list(title = paste(Aprefix, PCID_[1], '|', A2prefix, PCID_A2[1])),
    yaxis = list(title = paste(Aprefix, PCID_[2], '|', A2prefix, PCID_A2[2])),
    showlegend = TRUE
  )

# Print the plot
show(fig)
}

df_eth<-read.table('psam.tmp',sep='\t',header=TRUE)
df_pc<-read.table('evec.tmp',sep='\t',header=TRUE)
df_merge<-merge(df_eth,df_pc,by=c('IID'))

library(plotly)

popstrat_plot2d(df_merge,c('PC1','PC2'),Aprefix='ETHNICITY')

```

We see some outliers here, so let's filter out!

```

outliers<-df_pc[df_pc$PC1< (-0.2),]
write.table(outliers$IID,file='results/qc/qc3/pc1_outliers.txt',quote=F,row.names=F,col.names=F)

```

Run second pass of Principal Components:

```
plink2 --pfile results/qc/qc3/PEG_PD_LE --remove results/qc/qc3/pc1_outliers.txt --freq counts --pca bi
sed 's/^#//' results/qc/qc3/PEG_PD_pca2.eigenvec > evec.tmp
```

Generate new version with samples filtered out.

```
plink2 --pfile results/qc/qc2/PEG_PD_IMPUTED --remove results/qc/qc3/pc1_outliers.txt --make-pgen --out
```

Chunk into chromosomes:

```
for i in {1..22}; do
    plink2 --pfile results/qc/qc3/PEG_PD_IMPUTED2 --chr $i --export Av --out chunks/PEG_PD_CHR_$i
done
```

Prepare for GWAS:

In R:

```
df_covar<-read.table('covar.txt',header=T)
df_eth<-read.table('evec.tmp',header=T,sep='\t')
df_merge<-merge(df_covar,df_eth,by=c('IID'))
write.table(df_merge,file='covar_evec.txt',row.names=F,quote=F)
```

Run GWAS:

```
plink2 --pfile results/qc/qc3/PEG_PD_IMPUTED2 --glm omit-ref hide-covar no-firth --pheno-name DISEASE --
sed 's/^#//' results/gwas/PEG_PD.DISEASE.glm.logistic > ../GWAS_meta/gwas_peg.txt
```

Prep for meta-analysis in R:

```
# go to rawdata dir
setwd('..')
df_peg<-read.table('GWAS_meta/gwas.txt',sep='\t',header=T)
df_nalls<-read.table('..Nalls_GWAS/PDNallsP005.csv',header=T,sep=',')
df_nalls$ID<-paste(df_nalls$chr,df_nalls$position,sep=':')
df_nalls$N<-df_nalls$ncase+df_nalls$ncontrol
write.csv(df_nalls,file='../GWAS_meta/gwas_nalls.txt',row.names=F,quote=F)
df_merged<-merge(df_peg,df_nalls,by.x=c('CHROM','POS'),by.y=c('chr','position'))
```

Run meta in [repo_root]/rawdata/GWAS_meta:

```
cd ../GWAS_meta
../metal < metal.script
#Sort:
df_meta<-read.table('GWAS_OUT1.metal',sep='\t',header=T)
write.table(ordered,file='sorted_meta.txt',quote=F,row.names=F)
```